Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

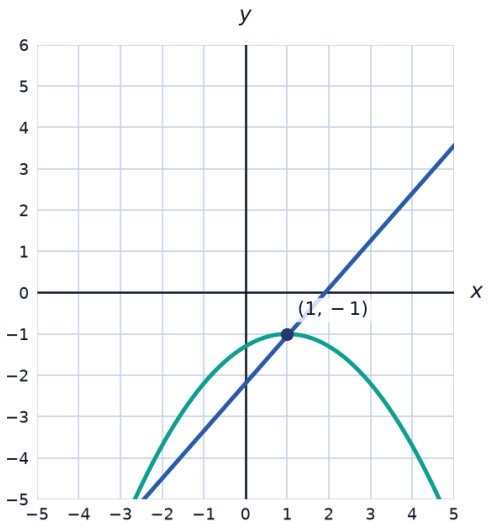
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $A(1, 3)$ and $B(7, 11)$.

2 **[Foundational]** • **Target: Distance and midpoint**

Find the midpoint of the segment joining $(-3, 5)$ and $(9, -1)$.



3 **[Foundational]** • **Target: Distance and midpoint**

The midpoint of $A(2, -3)$ and $B(x, y)$ is $(5, 1)$. Find the coordinates of B.

4 **[Foundational]** • **Target: Gradient and equations of lines**

Find the gradient of the line through $(2, -1)$ and $(6, 7)$.

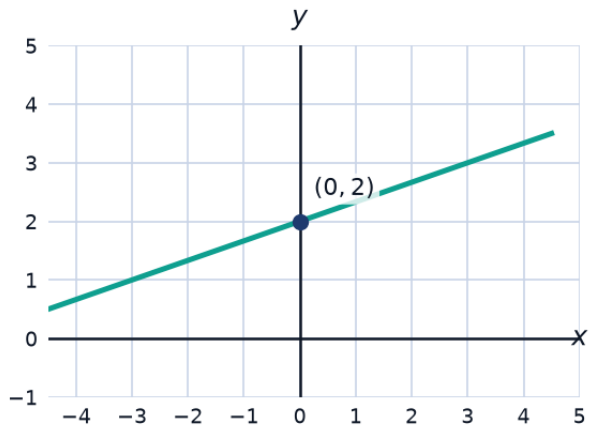
5 **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line parallel to $y = 3x - 1$ passing through $(5, 0)$.

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6 [Foundational] • Target: Parallel and perpendicular lines

Find the equation of the line perpendicular to $y = \frac{1}{3}x + 2$ passing through $(6, -1)$.



7 [Foundational] • Target: Parallel and perpendicular lines

Determine whether the lines $y = 4x - 3$ and $4y + x = 8$ are parallel, perpendicular, or neither.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The equation of a circle

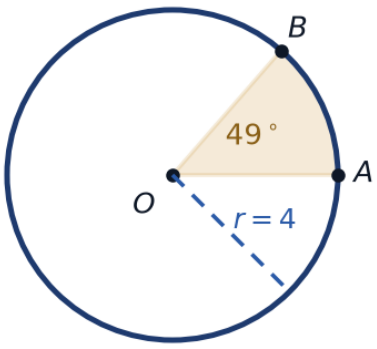
Write the equation of a circle with center $(3, -2)$ and radius 6.

9 [Intermediate] • Target: The equation of a circle

Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.

10 [Intermediate] • Target: The equation of a circle

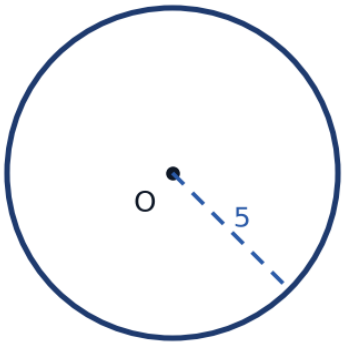
A circle has equation $(x + 1)^2 + (y - 4)^2 = 49$. State its center and radius.



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11 [Intermediate] • Target: Lines and circles together

A circle has center $(0, 0)$ and radius 5. Determine whether the line $y = x + 8$ intersects the circle, is tangent to it, or misses it entirely, by comparing the distance from the center to the line against the radius.



12 [Intermediate] • Target: Lines and circles together

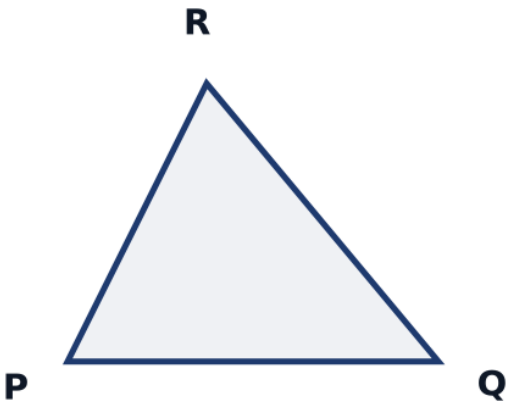
A circle has center $(2, 1)$ and radius 3. Determine whether the line $x + y = 10$ intersects the circle, is tangent to it, or misses it entirely.

13 [Intermediate] • Target: The perpendicular bisector

Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(7, 8)$.

14 [Intermediate] • Target: Area of a triangle from coordinates

Find the area of the triangle with vertices $P(0, 0)$, $Q(8, 0)$, $R(3, 6)$.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 18.01 Binomial Distribution

This question has two parts. Two houses are located at $A(0, 0)$ and $B(12, 0)$ (positions in kilometers). A cell tower must be built equidistant from both houses, and it must also lie on the line $y = x + 2$. (a) Find the equation of the perpendicular bisector of AB, and use it together with $y = x + 2$ to find the tower's coordinates. (b) Find the distance from the tower to each house, and confirm they are equal.

16 [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 19.01 Normal Distribution

[Geometry & Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

17 [Challenging] • Target: Distance and midpoint • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Gradient and equations of lines • Cross-Lesson: 3.01 Graphs & Transformations

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 5.01 Trigonometry

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 [Challenging] • Target: The equation of a circle • Cross-Lesson: 6.01 Differentiation & Integration

[Trigonometry & Triangle Laws · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distance and midpoint

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. 25

b. 50

c. 100

d. 125

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22 [Foundational] • Target: Gradient and equations of lines

A sector OAB of a circle with centre O and radius 12 cm has angle $\angle AOB = \frac{\pi}{2}$ radians. The point C is the midpoint of the radius OA. The line passing through C perpendicular to OA intersects the circular arc AB at the point D. The straight line segment BC is drawn. Find the exact area, in cm^2, of the region bounded by the line segment CD, the circular arc BD, and the line segment BC.

a. $12\pi + 18\sqrt{3} - 36$

b. $24\pi + 18\sqrt{3} - 36$

c. $12\pi + 36\sqrt{3} - 36$

d. $18\pi + 18\sqrt{3} - 36$

23 [Foundational] • Target: Parallel and perpendicular lines

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. 4

c. 5

d. 6

24 [Foundational] • Target: The equation of a circle

In the triangle ABC shown, point D lies on AB and point E lies on AC such that DE is parallel to BC. Given that AD = 6, DB = 4, and the area of triangle ADE is 27 square units, find the area of triangle ABC.

a. 45

b. 60

c. 75

d. 90

25 [Foundational] • Target: Lines and circles together

The locus $|z - (-6 + i)| = |z - (2 - 2i)|$ is the perpendicular bisector of the segment joining the points representing z_1 and z_2. Find the x-coordinate of the midpoint of this segment.

a. $-\frac{1}{2}$

b. 2

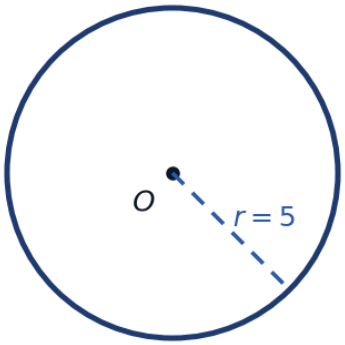
c. -2

d. -6

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26 [Foundational] • Target: The perpendicular bisector

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.



a. 25

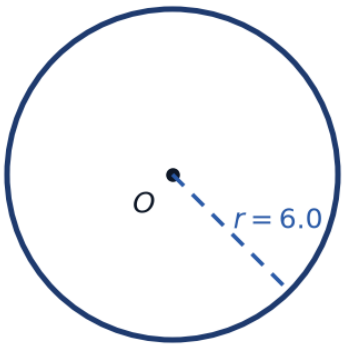
b. 30

c. 35

d. 40

27 [Foundational] • Target: Area of a triangle from coordinates

A sector of a circle with radius 6 cm has area 27 cm^2. Find the angle of the sector, in radians.



a. 1.5

b. 3

c. 0.75

d. 2

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28 [Foundational] • Target: A well-designed word problem: locating a cell tower

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2

b. 2

c. -4

d. 4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Distance and midpoint

In triangle PQR, angle PQR = 90 degrees. S is a point on PR such that QS is perpendicular to PR. The length of PQ is 15 cm and the length of QR is 20 cm. Let the length of PS be x cm and the length of SR be y cm. Find the value of x.

a. 10 cm

b. 12 cm

c. 15 cm

d. 9 cm

30 [Intermediate] • Target: Gradient and equations of lines

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

31 [Intermediate] • Target: Parallel and perpendicular lines

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

a. 6

b. 8

c. 10

d. 4

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32 [Intermediate] • Target: The equation of a circle

A circle has its center at the origin $(0, 0)$ and passes through the point $P(3, y)$. If the equation of the circle is $x^2 + y^2 = 25$, what is the positive value of y?

a. 3

b. 4

c. 5

d. 16

33 [Intermediate] • Target: Lines and circles together

In the diagram, points A, B, and C form a triangle. Point D lies on AC such that BD is the angle bisector of angle ABC. Given that AB = 10 cm, AC = 15 cm, and BC = 12 cm, and that AD = 6 cm, find the length of DC.

a. 8 cm

b. 7.2 cm

c. 9 cm

d. 6 cm

34 [Intermediate] • Target: The perpendicular bisector

In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.

a. 5 cm

b. 6 cm

c. 7 cm

d. 3 cm

35 [Intermediate] • Target: Area of a triangle from coordinates

In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. 9

b. 10

c. 15

d. 22.5

36 [Intermediate] • Target: A well-designed word problem: locating a cell tower

In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. 7

b. 9

c. 10

d. 12

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37 [Intermediate] • Target: Distance and midpoint

A circle is centered at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the equation of this circle?

a. $x^2 + y^2 = 7$

b. $x^2 + y^2 = 9$

c. $x^2 + y^2 = 16$

d. $x^2 + y^2 = 25$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Gradient and equations of lines • Cross-Lesson: 3.01 Graphs & Transformations

What is the radius of the circle with the equation $(x - 3)^2 + (y + 5)^2 = 36$?

a. 6

b. 36

c. 3

d. 5

39 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 5.01 Trigonometry

The area of a sector is 3 times its arc length for a circle of radius r. Find r.

a. 6

b. 3

c. 12

d. 9

40 [Challenging] • Target: The equation of a circle • Cross-Lesson: 6.01 Differentiation & Integration

Points $A(1, 0)$ and $B(7, -6)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

a. -1

b. 0

c. 2

d. 1

41 [Challenging] • Target: Lines and circles together • Cross-Lesson: 7.01 Statistics & Probability

A sector of a circle has radius 9 cm and angle 45°. Find the exact area of the sector.

a. $\frac{81\pi}{8}$

b. $\frac{49\pi}{8}$

c. 8π

d. $\frac{9\pi}{2}$

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42 [Challenging] • Target: The perpendicular bisector • Cross-Lesson: 8.01 Functions: Composite & Inverse

The polar equation $r = 2\cos\theta$ represents a circle. What is its radius?

a. 2

b. 3

c. 5

d. 1

43 [Challenging] • Target: Area of a triangle from coordinates • Cross-Lesson: 9.01 Circular Measure

In triangle ABC, $a = 6$, $b = 8$, and $C = 90^\circ$. Find the exact area of the triangle.

a. 48

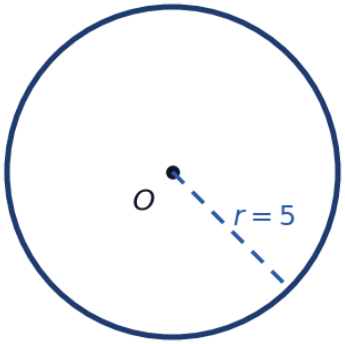
b. $24\sqrt{2}$

c. 24

d. $24\sqrt{3}$

44 [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

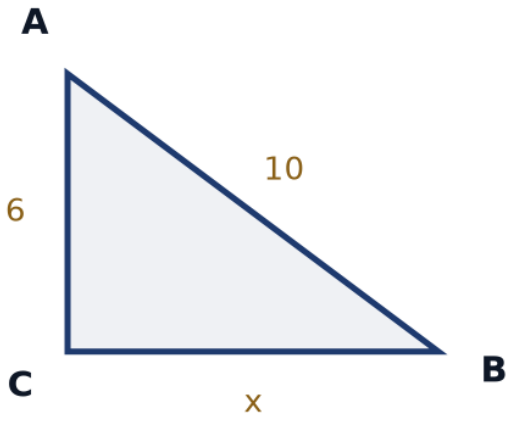
c. 1

d. $\frac{\sqrt{2}}{2}$

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45 [Challenging] • Target: Distance and midpoint • Cross-Lesson: 10.01 Series & Binomial Theorem

In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.



a. 10

b. 6

c. 8

d. 16

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